

Nihar Majalika

CCNA-certified networking · infrastructure · security · automation

linkedin.com/in/nihar-majalika-1bb75b199 · github.com/NiharMajalika

PROFESSIONAL PROFILE

CCNA-certified networking and infrastructure practitioner with postgraduate engineering work in telecommunications and networking. Documented projects span routing and switching, self-hosted infrastructure, network simulation, secure access, Linux administration, monitoring, backup automation and CI/CD.

SELECTED TECHNICAL PROJECTS

Personal infrastructure, monitoring and CI/CD homelab

[Proxmox](#) · [Ubuntu Server](#) · [Docker Compose](#) · [Pi-hole](#) · [Uptime Kuma](#) · [Nextcloud](#) · [Tailscale](#)

- Built a Proxmox-hosted Ubuntu Docker environment for DNS monitoring, availability checks, private cloud storage and container management.
- Implemented private remote access, weekly backups of application data and configuration, and a monitored backup-health signal.
- Documented a GitHub Actions deployment path using Tailscale, SSH and Docker Compose for a containerised homelab status application.

GIS-assisted hybrid communications and segmented OT research

[Master's thesis project](#) · [QGIS](#) · [ns-3](#) · [Cisco Packet Tracer](#) · [C++](#) · [Python](#)

- Designed and evaluated a hybrid communication architecture for a lagoon-based wastewater treatment plant case study.
- Modelled 423 infrastructure nodes and 395 communication links in QGIS, then evaluated packet delivery, delay, throughput and loss in simulation.
- Validated VLAN separation, redundant core paths, gateway routing, an OT DMZ and restricted firewall flows in Packet Tracer.

Network Systems and Web Security laboratory portfolio

[CSE5NSW](#) · [10 documented simulation and applied-cryptography labs](#)

- Configured and analysed DNS, HTTP/HTTPS, FTP, email, NTP and AAA services in Cisco Packet Tracer.
- Compared Telnet and SSH traffic, implemented standard and extended ACLs, and analysed a remote-access IPsec VPN path.
- Explored hashing, signing, certificate verification and PGP/GPG encryption, then applied access controls to simulated cloud and IoT scenarios.

CERTIFICATION

Cisco Certified Network Associate (CCNA 200-301)

CORE TECHNICAL SKILLS

Networking: TCP/IP, routing, switching, VLANs, OSPF, DNS, WAN concepts, packet analysis and troubleshooting

Security and access: SSH, ACLs, IPsec, WireGuard/Tailscale, firewall configuration, TLS, certificates and PKI concepts

Infrastructure: Linux administration, Proxmox, KVM/QEMU, Docker, Docker Compose, Portainer, Nextcloud and MariaDB

Automation and analysis: Bash, cron, GitHub Actions, CI/CD, Python, C++, QGIS, ns-3, FlowMonitor, NetAnim, pandas and matplotlib

EDUCATION

Master of Engineering - Telecommunications and Networking

La Trobe University, Bundoora VIC · Jul 2024 - Jun 2026 (dates from supplied resume)

- Project work includes telecommunications planning, network simulation, segmented operational technology architecture and network security.

Bachelor of Engineering - Electronics and Communication Engineering

N.M.A.M. Institute of Technology, Nitte, India · Graduated Jun 2023

EXPERIENCE

Management and Site Intern

Goa Shipyard Ltd, Vasco da Gama, India · Jul 2022 - Aug 2022

- Supported equipment inspections, calibration checks and field observations, then documented technical issues and maintenance findings.
- Assisted engineering staff with electronic and electrical equipment checks, including pressure gauges and EM-log systems.

Container Unloader - Casual on-hire

HireGalaxy Personnel · Host: Viewscape, Bayswater VIC · Nov 2025 - Jun 2026

- Worked safely in a fast-paced warehouse environment while following site induction, WHS, PPE and manual-handling requirements.

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